

■ Power Query Data Cleaning and Transformation Process

Dataset: Sales, Customers, Products, Stores, Exchange Rates

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Tool: Microsoft Power BI – Power Query Editor

1. Data Import and Initial Review

Imported five source tables: Sales, Customers, Products, Stores, and Exchange Rates. Inspected column data types, headers, and null values using Column Quality and Column Distribution in Power Query. Detected initial issues: inconsistent date formats, currency symbols, inconsistent naming, and missing delivery dates.

Rationale: Ensuring all fields have correct types and consistent formatting is essential for accurate relationships, DAX calculations, and visuals later in the model.

2. Fixing Date and Time Data Types

Changed data types for all date columns (Order Date, Delivery Date, Open Date, Date in Exchange Rates) to Date type. Handled the day-month vs. month-day format issue using locale conversion (Transform → Data Type → Using Locale → Date → English US). Removed invalid date rows and ensured consistent formatting.

M Code Snippet: `Table.TransformColumnTypes(#"Previous Step", {"Order Date", type date}, "en-US")`

Rationale: Date consistency ensures correct time intelligence calculations and proper relationships with the date table.

3. Cleaning Monetary Fields

Removed the \$ symbol from Unit Cost USD and Unit Price USD using "Replace Values." Changed data type to Decimal Number.

M Code Snippet: `Table.TransformColumns(#"Previous Step", {"Unit Price USD", each Number.FromText(Text.Replace(_, "$", "")), type number})`

Rationale: Converting to numeric values enables aggregation (SUM, AVG) and accurate financial measures.

4. Normalizing Tables and Reducing Redundancy

Normalized tables to reduce data redundancy and improve performance. Created separate dimension tables such as Category and Subcategory, removed corresponding columns from the Products table, and connected them using keys (CategoryKey, SubcategoryKey).

Rationale: Normalization reduces file size, improves query performance, and ensures a consistent snowflake schema.

5. Handling Missing Values and Derived Columns

Detected 86% missing values in Delivery Date. Business understanding confirmed that missing values occur for in-store purchases (no delivery). Created a new column Order Type: "Online" if StoreKey = 0, "In-store" otherwise.

M Code Snippet: `Table.AddColumn("#Previous Step", "Order Type", each if [StoreKey] = 0 then "Online" else "In-store")`

Rationale: Clarifies delivery-related nulls and provides an additional analytical dimension for sales type comparison.

6. Data Modeling and Schema Design

Designed a Snowflake Schema connecting Sales (fact table) to dimension tables: Customers, Products, Stores, Exchange Rates, and Date (created in Model View). Verified relationships (One-to-Many from dimension to fact) and ensured referential integrity (no orphan keys).

Rationale: Snowflake schema minimizes redundancy, improves query efficiency, and enhances analytical clarity.

7. Summary and Impact

Data is now fully standardized and ready for analysis. Model efficiency improved through normalization and relationship validation. Created a new analytical feature (Order Type) and established consistent date and currency handling for accurate reporting.